hmm

Saturday, 7 March 2026 9:49 AM

• - input

• - sigmoid neuron

Each neuron is a sigmoid

In L, in z_1 list

$$z_1 = wx_1 + w_{12}x_2 + w_{13}x_3 + b_{21}$$

in a_1 list

$$a_{11} = \frac{1}{1+e^{-z_1}}$$

$$a_{12} = \frac{1}{1+e^{-z_2}}$$

$$\vdots$$

For layer 2 in z_2 list

$$z_1 = w_{21}a_1 + w_{22}a_2 + w_{23}a_3 + w_{24}a_4 + w_{25}a_5 + w_{26}a_6 + b_{21}$$

in a_2 list

$$a_{21} = \frac{1}{1+e^{-z_1}}$$

$$\vdots$$

$$a_{24} = \frac{1}{1+e^{-z_4}}$$

For layer 3:

$$\boxed{z = w_{31}a_{11} + w_{32}a_{12} + w_{33}a_{13} + w_{34}a_{14} + b_{31}}$$

$$a_{31} = \frac{1}{1+e^{-z}}$$

$$\boxed{\hat{y} = a_3}$$

$$loss = \frac{1}{2}(y - \hat{y})^2$$

$$\delta_3 = \frac{dC}{dz^3} = \frac{\partial C}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z}$$

$$\delta_3 = \frac{\partial C}{\partial z^3} = (\hat{y} - y)\sigma(z)(1-\sigma(z))$$

$$\hat{y} = \frac{1}{1+e^{-z}}$$

$$\frac{d\hat{y}}{dz} = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\frac{\partial \hat{y}}{\partial z} = \sigma(z)(1-\sigma(z))$$

$$\delta_2 = \frac{\partial C}{\partial z^2} = \frac{\partial C}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_3} \cdot \frac{\partial z_3}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_2}$$

$$\delta_2 = \delta_3 \cdot \frac{\partial z_3}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_2}$$

In L-1 δ_2

$$\delta_2 = \delta_3 \cdot \frac{\partial z}{\partial a_{31}} \cdot \frac{\partial a_{31}}{\partial z_2}$$

$$z_3 = w_{31}a_1 + w_{32}a_2 + w_{33}a_3 + w_{34}a_4 + b_{31}$$

$$\frac{\partial z_3}{\partial a_1} = w_{31}$$

$$\frac{\partial z_3}{\partial a_2} = w_{32}$$

$$\vdots$$

$$\delta_2 = \delta_3 \cdot w_{31}(a_{31})(1-a_{31})$$

$$a_1 = \frac{1}{1+e^{-z_1}}$$

$$\frac{\partial a_1}{\partial z_1} = \frac{-e^{-z_1}}{(1+e^{-z_1})^2}$$

$$= \sigma(z_1)(1-\sigma(z_1))$$

$$\frac{\partial a_2}{\partial z_2} = a_2(1-a_2)$$

$$\frac{\partial C}{\partial w_3} = \left[\frac{\partial C}{\partial \hat{y}} \cdot \frac{\partial a_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1}\right]$$

$$\frac{\partial C}{\partial w_1} = \delta_2 \cdot \frac{\partial z_1}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1}$$

$$= \delta_2 \cdot \frac{\partial z_{26}}{\partial a_{16}} \cdot \frac{\partial a_{16}}{\partial z_1}$$

$$\delta_1 = \delta_2 \cdot w_{26} \cdot a_1(1-a_1)$$

$$z_{26} = w_{21}a_1 + w_{22}a_2 + w_{23}a_3 + w_{24}a_4 + w_{25}a_5 + w_{26}a_6$$

$$\frac{\partial z_{26}}{\partial a_1} = w_{21}$$

$$\vdots$$

$$\frac{\partial z_{26}}{\partial a_{16}} = w_{26}$$

$$a_1 = \frac{1}{1+e^{-z_1}}$$

$$\frac{\partial a_1}{\partial z_1} = a_1(1-a_1)$$

$$w_3 = w_3 - \alpha \cdot \frac{\partial C}{\partial w_3}$$

$$\frac{\partial C}{\partial w_3} = \frac{\partial C}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_3}$$

$$\frac{\partial C}{\partial w_3} = \delta_3 \cdot a_{31}$$

$$w_3 = (\delta_3 \cdot a)(\alpha \cdot 1)$$

$$w_2 = (\delta_2 \cdot a_1)(\alpha \cdot 1)$$

$$w_1 = (\delta_1 \cdot x)(\alpha \cdot 1)$$